

Zihan Yang

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About Me

I am a master's student at Beihang University with a background in materials science and aerospace engineering, now focusing on robotics. My research interests lie at the intersection of control and machine learning, covering both learning-based planning and control, as well as control-theoretic methods that improve learning algorithms. Recent works have been recognized at ICLR 2025 (Oral) and ICCA 2024 (Best Student Paper Award).

Education

Northwestern Polytechnical University , B.Eng. in Material Science and Engineering	Sept 2019 - June 2023
Queen Mary University of London B.Eng. in Material Science and Engineering (UK-China Transnational Education)	Sept 2019 - June 2023
- GPA: 3.6/4.0 First Class Honors.	
Beihang University , M.S. in Aerospace Science and Technology	Sept 2023 - June 2026 (expected)
GPA: 3.8/4.0, supervised by Prof. Kexin Guo, Prof. Xiang Yu, and Prof. Lei Guo	

Publications

(* equal contribution)

- [1] **Z. Yang**, J. Jia, M. Wang, Y. Liu, K. Guo, and X. Yu, "Towards non-structural disturbance prediction: Feedback-calibrated meta-adaptation," *Under Review*, 2025.
- [2] J. Jia*, M. Wang*, **Z. Yang**, B. Yang, Y. Liu, K. Guo, and X. Yu, "Learning-based observer for coupled disturbance," 2025.
- [3] J. Jia*, **Z. Yang***, M. Wang, K. Guo, J. Yang, X. Yu, and L. Guo, "Feedback favors the generalization of neural ODEs," in *The Thirteenth International Conference on Learning Representations*, **Oral Presentation (acceptance rate: 1.8%)**, 2025.
- [4] K. Guo*, **Z. Yang***, J. Jia, Y. Liu, and X. Yu, "Optimizing control-friendly trajectories with self-supervised residual learning," *Under Review*, 2024.
- [5] **Z. Yang**, J. Jia, Y. Liu, K. Guo, X. Yu, and L. Guo, "TRACE: Trajectory refinement with control error enables safe and accurate maneuvers," in *IEEE 18th International Conference on Control and Automation*, **Best Student Paper Award**, 2024.
- [6] X. Song, X. Guo, G. Liu, **Z. Yang**, and L. Liu, "Co-optimization of motion and energy domain for hydrogen-powered hybrid UAVs: A bi-directional coupling architecture," in *IEEE 19th International Conference on Control and Automation*, 2025.
- [7] K. Guo, Y. Liu, J. Jia, **Z. Yang**, S. Zhou, X. Yu, and L. Guo, "Flying in harsh environments: Anti-rain quantification system and control strategy," *IEEE Transactions on Industrial Electronics*, 2025.

Talks

Presentation <i>IEEE International Conference on Control and Automation (ICCA)</i> , Reykjavík, Iceland.	June, 2024
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Honors and Awards

Oral Presentation, Top 1.8% <i>The Thirteenth International Conference on Learning Representations (ICLR)</i> , Singapore.	April, 2025
Best Student Paper Award <i>IEEE International Conference on Control and Automation (ICCA)</i> , Reykjavík, Iceland.	June, 2024

Funds

Second-Class Scholarship Beihang University, Beijing, China.	2024
Second-Class Scholarship Beihang University, Beijing, China.	2023

Skills

Coding: Python (skilled), Matlab/Simulink (skilled), C/C++ (familiar), HTML/CSS (basic)

Hardware: Skilled in UAV hardware-software co-design and system integration; developed fixed-wing and quadrotor platforms with embedded systems (Pixhawk, Jetson Nano, Intel NUC); experienced in rapid prototyping (3D printing, laser cutting), electronic assembly (soldering)...

Language: Mandarin Chinese (native), English (fluent), Japanese (basic)

Misc.: Electric guitar, skiing, badminton, swimming...

Soft Skills.: Self-motivation, critical-thinking, multitasking...